

# Brock Allan

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## Education

### University of California, Irvine

Bachelor of Science in **Computer Science**, GPA 3.6

- **Coursework:** Data Structures & Algorithms, Security & Privacy, Information Retrieval, System Design, Operating Systems, **Computational Complexity**, Differential Equations, Machine Learning
- **Distinctions:** Campuswide Honors Collegium, Published AI & Game Design Thesis, Dean's Honor List, CLAP certificate, Chancellors Excellence, FFA President, NHS President

## Skills & Tools

- **Coding Languages:** Python, Java, JavaScript, C, C++, C#, Lisp, Prolog, HTML, **Assembly**, R
- **Industry Software:** Blender, Unity, **Unreal Engine**, Photoshop, **Vim**, **Git**, Spreadsheets, **GDB**

## Technical Experience

### Limitations of AI & LLMs in Game Design Methodology – Thesis

September 2025 – June 2026

- Led Advanced AI Research & Agile Workflows via acting as the Product Owner under the **Scrum framework**. Led a multi-stage empirical study evaluating the capabilities and structural constraints of state-of-the-art LLMs (including **GPT-4.3** and **GPT-5.5**), managing iterative sprints, analyzing critical LLM anomalies like **context drift** and **token-level optimization**, and adapted pipelines across sequential interactions.
- Designed and evaluated complex natural language programming strategies ranging from **Zero-Shot** and **Few-Shot** to **Simulated Tuning (Chain-of-Hindsight and Self-Correction paradigms)**, to optimize code quality, reduce intrinsic and extrinsic hallucinations, and mitigate downstream algorithmic defects in generative systems.
- Codified a multi-dimensional quantitative rubric mapping 6 distinct variables (Clarity, Flow, Balance, Length, Integration, Fun) onto a **1–7 scale** to empirically benchmark human-AI interaction. Isolated critical performance deltas demonstrating that guided context-aligned prompt engineering raised user experience metrics to a peak score of **6.33 / 7.0**.

### The Dark Lord Beckons

July 2025 - Current

- Created a **3D-2D Rogue-Lite RPG** from scratch using self made **Unity** assets and physics.
- Developed original assets, scripts, lighting, **animations**, and the environment in Unity. Scripts and animations were coded in **C#**.
- Developed **functional** game components and iteratively improved game **design**.

### Rocky Lagoon

April 2025 - June 2025

- Created an **immersive** VR pirate-themed game inspired by interactive Disneyland rides.
- Developed original assets, scripts, lighting, animations, and models in Unity in **C#**.
- Gained hands-on expertise in producing **high-quality game assets** and publishing polished **VR** games.

## Professional Experience

### Learning Assistant, Intro to Programming, Shannon Alfaro, UCI

September 2025 - Current

- Instructed College students in how to code and design programs to achieve **foundational proficiency in core computing concepts**.
- Closely guided two groups of 6 students through technical coursework, hosted regular **office hours** to clarify complex topics and provided **detailed feedback** on assignments.
- Directed the LA team to **streamline communication**, provided **actionable feedback** to course instructors, and served as the **primary student advocate** to drive course improvements.

### Freelance Game Developer

June 2024 - Current

- Utilized game engines such as **Unity** and **Unreal Engine 5**, programming in **C++** and **C#**, to create immersive games in **VR, 2D, 3D, and FPS genres**.
- Designed gameplay systems, developed assets, and implemented interactive mechanics from the ground up for projects such as “Rocky Lagoon.”